

Learning goals:

- Understand logistic regression and how to implement and apply it;
- Understand the differences between logistic regression and linear discriminant analysis;
- Understand the probabilistic output of logistic regression;
- Understand the implications of non-convex cost functions;
- Understand gradient descent (for linear regression).

Question 1: Refresh your understanding

Refresh your understanding in the following concepts:

- Loss function(s): IBM
- Gradient descent: IBM, Geeks for Geeks

Question 2: Properties of logistic regression

- Write down the objective function of logistic regression.
- For each variable in the objective function, provide the dimensionality and the conceptual meaning.
- Does logistic regression have an analytical solution?
- Is logistic regression a supervised or unsupervised method?
- Is logistic regression used for classification or for regression?

Question 3: Objective function of logistic regression

The following cost function for logistic regression needs to get minimized to determine the optimal choice of the parameters $\mathbf{w}$:

$$J_1(\mathbf{w}) = \sum_{i=1}^N \ln(\exp(-\mathbf{y}_i(\mathbf{w}^T \mathbf{x}_i)) + 1)$$

- Assuming linear separability of a given dataset, what can happen to the weights $\mathbf{w}$ when minimizing the cost function $J_1(\mathbf{w})$? Why is this a problem?

Question 4: Regularization of logistic regression

A more stable estimate for $\mathbf{w}$ can be achieved when using the concept of **regularization**. Adding an additional term $J_2(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T \mathbf{w}$ to the cost function modifies the overall cost function $J(\mathbf{w}, C)$ to:

$$J(\mathbf{w}, C) = J_1(\mathbf{w}, C) + J_2(\mathbf{w}) = C \cdot \sum_{i=1}^N \ln(\exp(-\mathbf{y}(\mathbf{w}^T \mathbf{X})) + 1) + \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

Note, that the so-called regularization parameter C determines the relation of the second cost function $J_2(\mathbf{w})$ with respect to the first cost function $J_1(\mathbf{w})$.

- What is the role of the additional cost function $J_2(\mathbf{w})$?

Question 5: Jupyter notebook logistic regression with gradient descent

Now put the theory into practice by going through the **logistic regression with gradient descent** Jupyter notebook.

Question 6: Jupyter notebook logistic regression

Now put the theory into practice by going through the **logistic regression** Jupyter notebook.

Question 7: Submitting to Brightspace

For this assignment, please submit the following:

- The two Jupyter notebooks with your solutions
- A report (pdf extension) with the answers to the theoretical questions. In this report, add a line stating who contributed to what (please, adhere to the code of conduct). Additionally, please add a line on whether, where, and how you used GenAI for this assignment.

Question 8: (Optional) Test your understanding with real exam questions

(a) Logistic regression

- 2022-2023 Part 4
- 2023-2024 Part 4
- 2023-2024 resit Part 4
- 2024-2025 resit Part 5